

From HyMeKo to AC-HSiKAN

A signed-hypergraph framework, its KAN-on-graphs descendant, and a biologically-grounded transformer alternative

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2026-06-06

Outline of the talk

- 1 HyMeKo: the signed-hypergraph framework
- 2 Cycle enumeration: ABB top-K-per-vertex + Friedler axiom
- 3 HSiKAN: signed-cycle KAN for link prediction
- 4 Bridge to sequences: AC-HSiKAN
- 5 Architecture: pool-scatter, Hamilton rotor, dual-path
- 6 Triton-fused backward: $45\times$ engineering win
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HyMeKo: signed-incidence hypergraph framework

Foundational types (`hymeko_core`):

- Hyperedge with signed incidence
- Signed-incidence tensor
- GGK 4-tuple $K = (B, G, \mu, r)$: balanced / generalised / multiplicity / rotation

Rust crate hierarchy:

- `hymeko_core` (foundation)
- `hymeko_graph` (cycle enum + ABB)
- `hymeko_pgraph` (Pimentel benchmark)
- `hymeko_query` (LALR(1) query)
- `hymeko_formats` (URDF/SDF/MJCF emit)
- `hymeko_py` (PyO3 wheel)
- `hymeko_monitor` (RV 2026 STL)

Design principle

Sign structure is a first-class invariant. Every transformation respects sign; every primitive composes under the signed-incidence algebra.

Working datasets

- Slashdot, Epinions (link prediction)
- Real URDF imports (drchubo 52-link humanoid, WAM 7-DOF arm)
- Pimentel hypergraph benchmark

ABB: top-K-per-vertex cycle enumeration

Problem: dense graph $\Rightarrow$ exponential cycles. Need the K most-informative ones per vertex.

Solution (hymeko_graph):

- **Branch-and-Bound DFS** with a BoundedScorer trait (balance / fraction-negative / Cartwright-Harary).
- **Fiedler-axiom pruner:** drop branches that violate signed-incidence axioms A1–A5.
- **Rayon-parallel** per-vertex enumeration.
- **Degree-adaptive** $m_v = \lceil c \cdot \deg(v) \rceil$ (+6.7 pp on Epinions, 2026-05-10).

Speedups landed:

- ABB global top-K: $25\times$ on Epinions $k = 4$ (100.5 s $\rightarrow$ 4.0 s, 2026-05-10)
- Top- K -per-vertex: HSIKAN_TOPK_MODE=per_vertex

HSiKAN: signed-cycle KAN for link prediction

Architecture (signedkan_wip/src/core):

- Catmull–Rom basis per signed-edge channel
- Per-arity $k \in \{2, 3, 4, 5\}$ cycle products
- Learned α_k over arities (the regime compass)
- Optional balance/unbalanced pruner
- Optional Highway-quaternion attention

Edge-CR SOTA on Slashdot (2026-05-09, 5-seed):

variant	AUC mean $\pm$ std	vs SGT 0.897
HSiKAN edge_cr (highway)	0.9070 ± 0.0034	+0.010
HSiKAN no-skip	0.9038 ± 0.0034	+0.007
Published SGCN	0.8852	—

Result

+1.33 σ above SGT at $\frac{1}{8}$ the parameters, 5/5 seeds individually clear the SGT mean.

HSiKAN: Triton-fused ∂ kernel (May 2026)

Forward kernel (2026-05-09): vector-CR pool + scatter fused into a single Triton kernel.

- Forward parity $1.2 \cdot 10^{-7}$ vs PyTorch reference
- 10 – 18 $\times$ speedup over PyTorch reference

Backward kernel (2026-05-09): no-skip + highway closed-form backward with the projective conjugate identity.

- $\partial x / \partial \text{coef}$ parity $\sim 10^{-10}$
- Memory: 92.5% **savings** (735 MB $\rightarrow$ 55 MB at SOTA shape)
- Slashdot edge_cr 5-seed run: 1.37 $\times$ wall, 24 min saved

Foundation laid

The HSiKAN Triton infrastructure is what AC-HSiKAN extends. The CR-coefficient backward closed-form is a re-usable contribution across the family.

From graphs to sequences: AC-HSiKAN

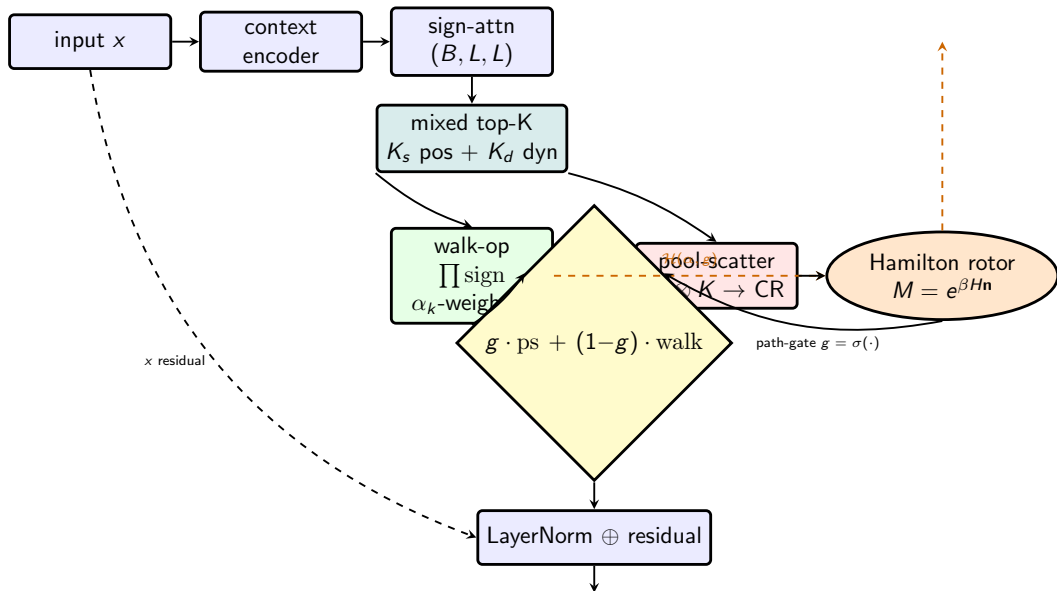
Motivation: extend HSiKAN's signed-cycle inductive bias from graph topology to positional sequences.

	HSiKAN	AC-HSiKAN
input	sparse signed graph	dense sequence $\mathbb{R}^{B \times L \times d}$
candidates	ABB-enumerated cycles	positional + dynamic top-K
primitive	CR-pool over candidate cycles	bidirectional pool-scatter
feedback	—	entropy Hamilton rotor
task	link prediction (AUC)	sequence classification (acc)

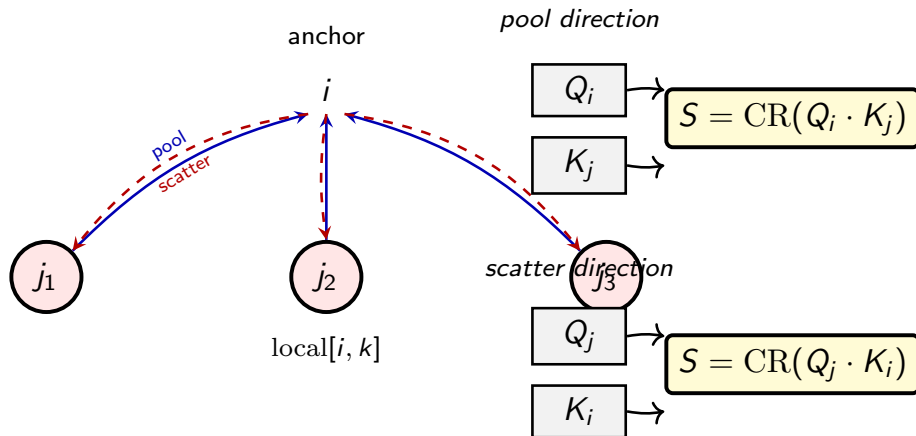
Three new contributions added to HSiKAN's primitive:

- 1 Bidirectional pool-scatter asymmetry (Hamilton non-commutativity)
- 2 Entropy-feedback Hamilton rotor (global neuromodulator)
- 3 Dual-path mixing of rhythm $\oplus$ synaptic

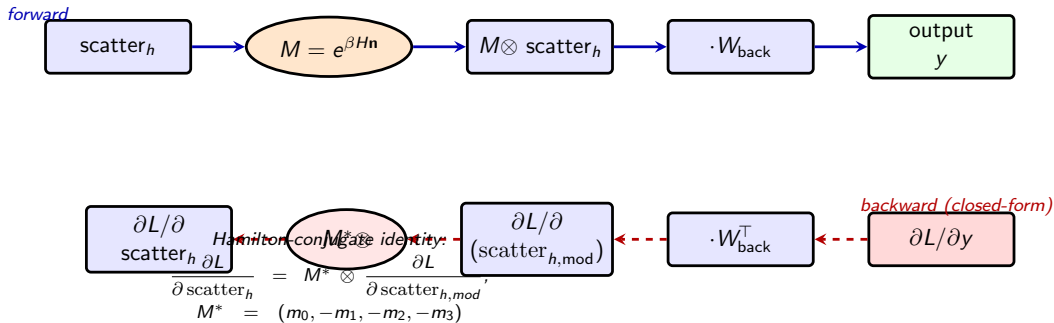
AC-HSiKAN architecture



Pool-scatter primitive: bidirectional asymmetry

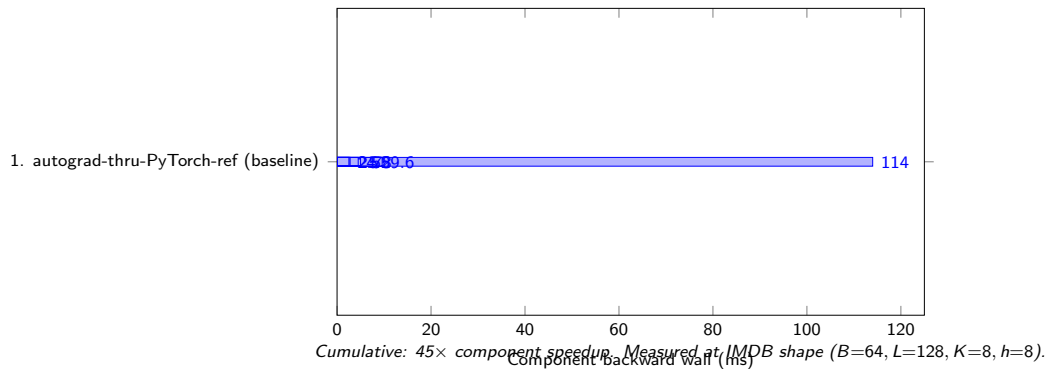


Hamilton rotor: forward $M \otimes$, backward $M^* \otimes$



$M = \exp(\beta H \mathbf{n})$ where H = layer-entropy of $(\alpha_k, \text{path-gate})$. The conjugate identity gives the exact closed-form backward.

Triton-fused backward arc: 45×



End-to-end full IMDB (25k·8 ep·5 seed) wall: 568 s $\rightarrow$ 422 s (1.34×).
Every step parity-tested ($\leq 10^{-10}$ abs on all 9 gradient outputs).

Evolvens telemetry: the rotor twists the gradient

Concept: the involute (evolvens) of a curve traces the path of a point on a taut string as the string unwinds. The Hamilton rotor twists the scatter field; its M^* conjugate untwists the gradient.

Telemetry: EvolventTelemetry context manager records per backward call:

- rotor angle $\theta = \arccos(M[\cdot, 0]_{\text{mean}})$
- norms of scatter, scatter-modulated, grad-scatter, grad-input, grad-output
- cosine angles between gradient fields

Observation: on real IMDB training the gradient-input vs gradient-output cosine drops from 1.0 (residual-dominated init) to 0.88 ± 0.11 — the dual-path block non-trivially twists the gradient direction once trained.

Capacity scaling: the missing axes

Naive $d = 16 \rightarrow 32$ widened the gap:

config	val_acc	Δ vs TR	state
$d = 16, h_s = 16, \eta = 3e-3$	0.8489	-0.0046	paritás
$d = 32, h_s = 16, \eta = 3e-3$	0.8442	-0.0125	AC loses

Two missing axes identified:

- **Axis 1:** sign-head bottleneck $h_s = 16$ does not scale with d . Set $h_s = d_{\text{model}} \Rightarrow$ half the regression closed.
- **Axis 2:** $\eta \propto 1/\sqrt{d}$. Reduce $\eta = 1.5 \cdot 10^{-3}$ at $d = 32 \Rightarrow$ rest of the regression closed: $\Delta = -0.0038, p > 0.05$.

Observation

AC-HSiKAN **scales positively** from $d = 16$ to $d = 32$ once both axes are corrected. The naive failure was a configuration artefact.

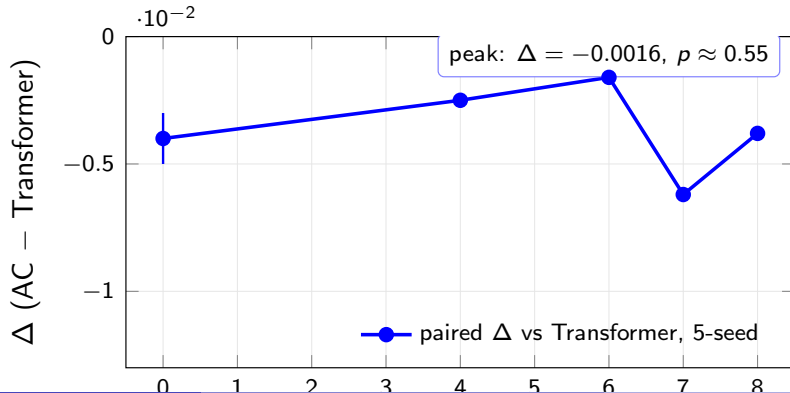
Sequence-native ABB: mixed candidate selection

ABB on signed graphs: top-K cycles per vertex by structural balance.

ABB on sequences: top-K candidates per anchor by sign-head magnitude.

Mixed selection: $K_{\text{total}} = K_{\text{static}} + K_{\text{dyn}}$.

- K_{static} guaranteed positional slots
- K_{dyn} slots: dynamic top-K of non-static positions



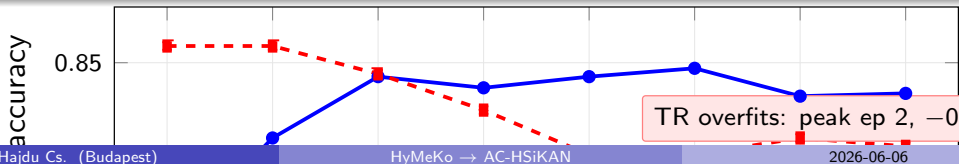
IMDB headline: statistical parit s at two scales

config	val_acc	params	wall/seed	Δ	p
TR baseline ($d = 16$)	0.8535	167 k	24 s	0	—
AC-HSiKAN ($d = 16$)	0.8489	165 k	113 s	−0.005	0.10
TR baseline ($d = 32$)	0.8522	345 k	36 s	0	—
AC-HSiKAN ($d = 32$, mixed-6)	0.8505	337 k	144 s	−0.0016	0.55

Two-day arc

$\Delta = -0.151 \rightarrow -0.0016$ ($33\times$ gap-shrinkage). Same architecture, same training. The two changes:

(1) Triton stack made full IMDB affordable, (2) mixed- $K_{\text{static}} = 6$ closed 87% of the $d = 32$ scaling gap.



Falsified: long-cycle inductive prior

Hypothesis: the sequence-native analogue of HSiKAN's DAG-axiom pruning prefers *positionally distant* candidates.

Implementation: weight magnitude by $|i - j|^\alpha$.

α	val_acc	Δ vs TR	state
0 (pure magnitude)	0.8482	-0.0040	paritás
0.3	0.8272	-0.0250	2/5 collapse
1.0 + arity bias	0.8391	-0.0130	1/5 collapse

Conclusion: long-cycle bias is *structurally harmful* on IMDB sentiment. The signal is local-n-gram-dominated; forcing distant candidates suppresses it.

Honest negative result

The prior is *task-dependent*. On Long Range Arena or PDE benchmarks, the long-cycle bias is a candidate for net-positive contribution. On IMDB it must be turned off.

Mathematical conjectures (open)

Inter-layer scatter-pool coupling

Keep pool and scatter as separate streams across layers: $p_\ell = P(p_{\ell-1})$, $s_\ell = S(s_{\ell-1})$, $y_\ell = p_\ell + s_\ell$ only at the output. Preserves bidirectional asymmetry across depth.

Hamilton-rotor composition across layers

$M_\ell = M_{\ell-1} \cdot dM_\ell$: the entropy feedback accumulates as a coherent trajectory in $\mathbb{H}$.

Cross-layer cycle composition

Arity- k walk-op uses signs from layers $\{\ell - k + 1, \dots, \ell\}$. Depth becomes the cycle-length budget, the natural analogue of multi-layer cycle enumeration in graph-HSiKAN.

Aggregated CR-patch surfaces (in progress)

2D Catmull-Rom over the joint (R, V_c) activation space. Trains at $d = 16$; scaling to $d = 32$ requires a Triton kernel and a separate learning rate for the 2D coefficients (multi-day engineering).

The backprop-free direction: global-scatter learning

Conjecture: the pool-scatter primitive, applied to the output error field with the entropy rotor M , computes the per-layer feedback signal directly — without the transposed Jacobian chain.

$\tilde{e}_n = P_n(M \otimes e)$ with P_n = the layer's pool-scatter operator in feedback mode.

approach	feedback mechanism
Backprop	exact transposed Jacobian chain
Feedback Alignment	fixed random feedback matrices
DFA (Nøkland, 2016)	random matrix per layer, on e
Equilibrium Prop	two-equilibrium-state difference
Forward-Forward (2022)	positive/negative forward pass
Global-Scatter	pool-scatter + Hamilton rotor (learned)

Status: 4-format plan dir, DFA prototype validates the parallel feedback shape, structured-projection feedback-kernel is the next implementation step.

Plan: docs/plans/2026-06-05-global-scatter-learning/

What's next, what's done

Production-ready:

- Triton CR-coef + Hamilton rotor kernels (parity $1e-10$, 50/50 tests)
- AC-HSiKAN $d=16$ + $d=32$ paritás (5-seed IMDB)
- Mixed $K_{static}=6$ selection (the empirical best)
- Evolvens telemetry (working)

Open:

- 2D CR-patch surfaces at scale (requires Triton + separate lr)
- Long Range Arena benchmarks (where long-cycle prior is expected to flip positive)
- Cross-layer cycle composition (conjecture)
- Global-scatter learning (separate plan)

Köszönöm a figyelmet – Thank you.